INTRODUCTION TO MINING ENGINEERING

13 MINERALS IN YEMEN

Dr. Adnan Abdulaziz Barahim
Associated professor of engineering geology

Prepared for
Mining Engineering Department
AL Arab University

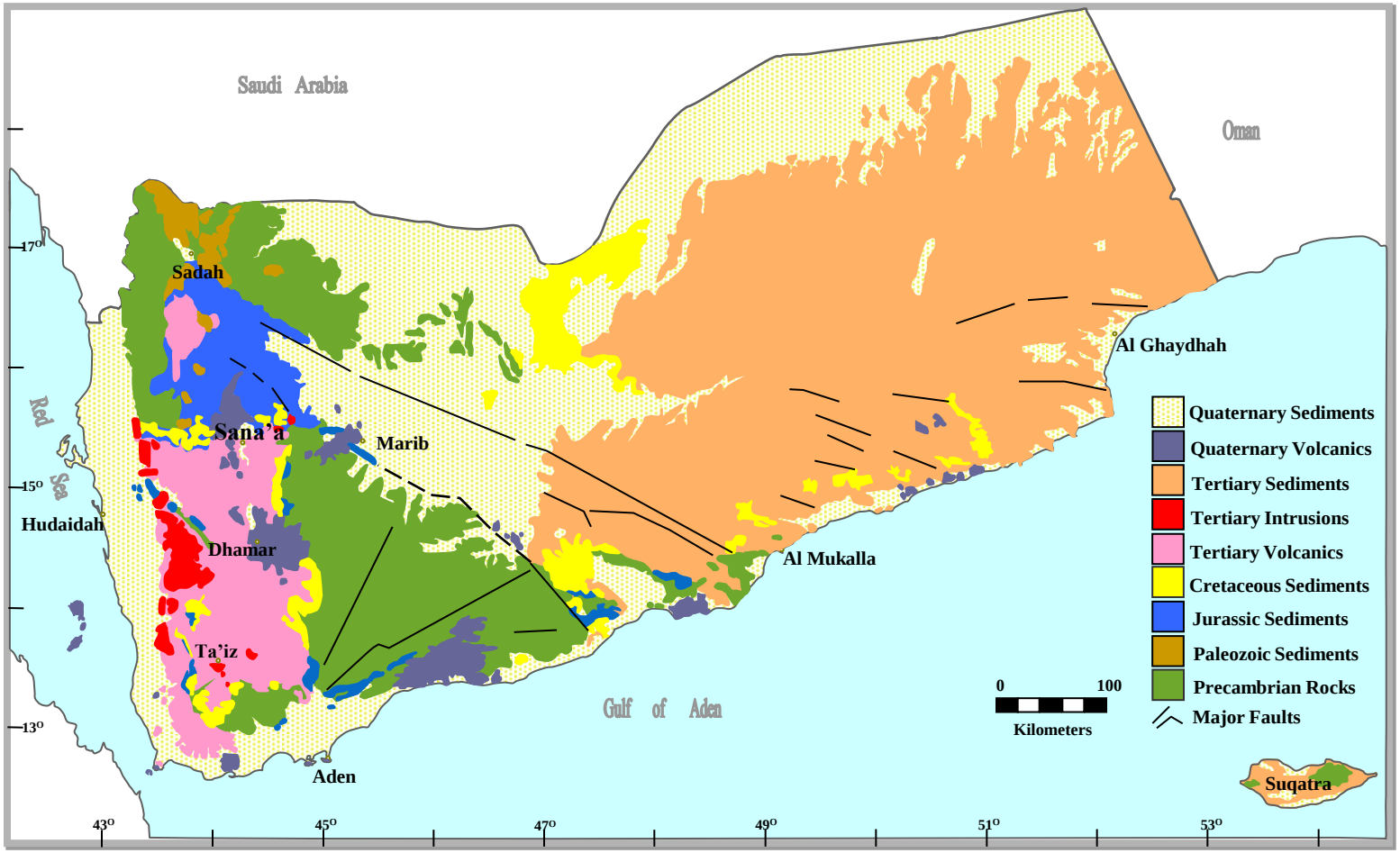
Minerals in Yemen

Topics

1. Geological map of Yemen
2. Authority responsible;
3. Distribution of Old mining
4. Yemen rich in minerals,
5. Infra structure;
6. Mining and quarry low, No. 22, 2010
7. Metallic minerals: Occurrence and its geology
8. Industrial minerals; Occurrence and its geology

Minerals in Yemen

- **Geological map of Yemen**

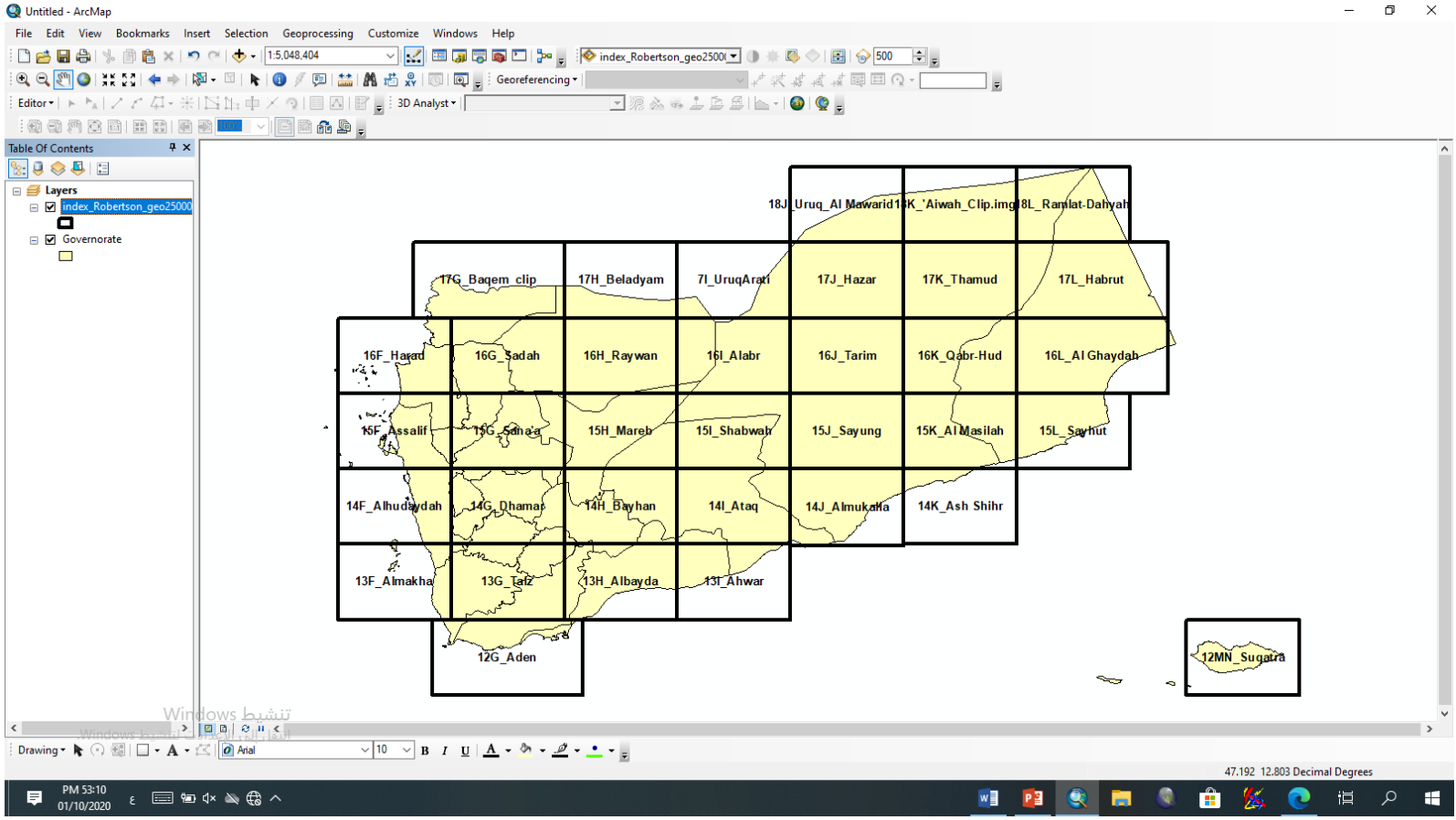


Minerals in Yemen

- **Authority responsible;** ؟
  - **Geological survey and mineral resources board** established in 1999 based on public ordinance no. 317
  - The history bac to 1970 oil and mineral board

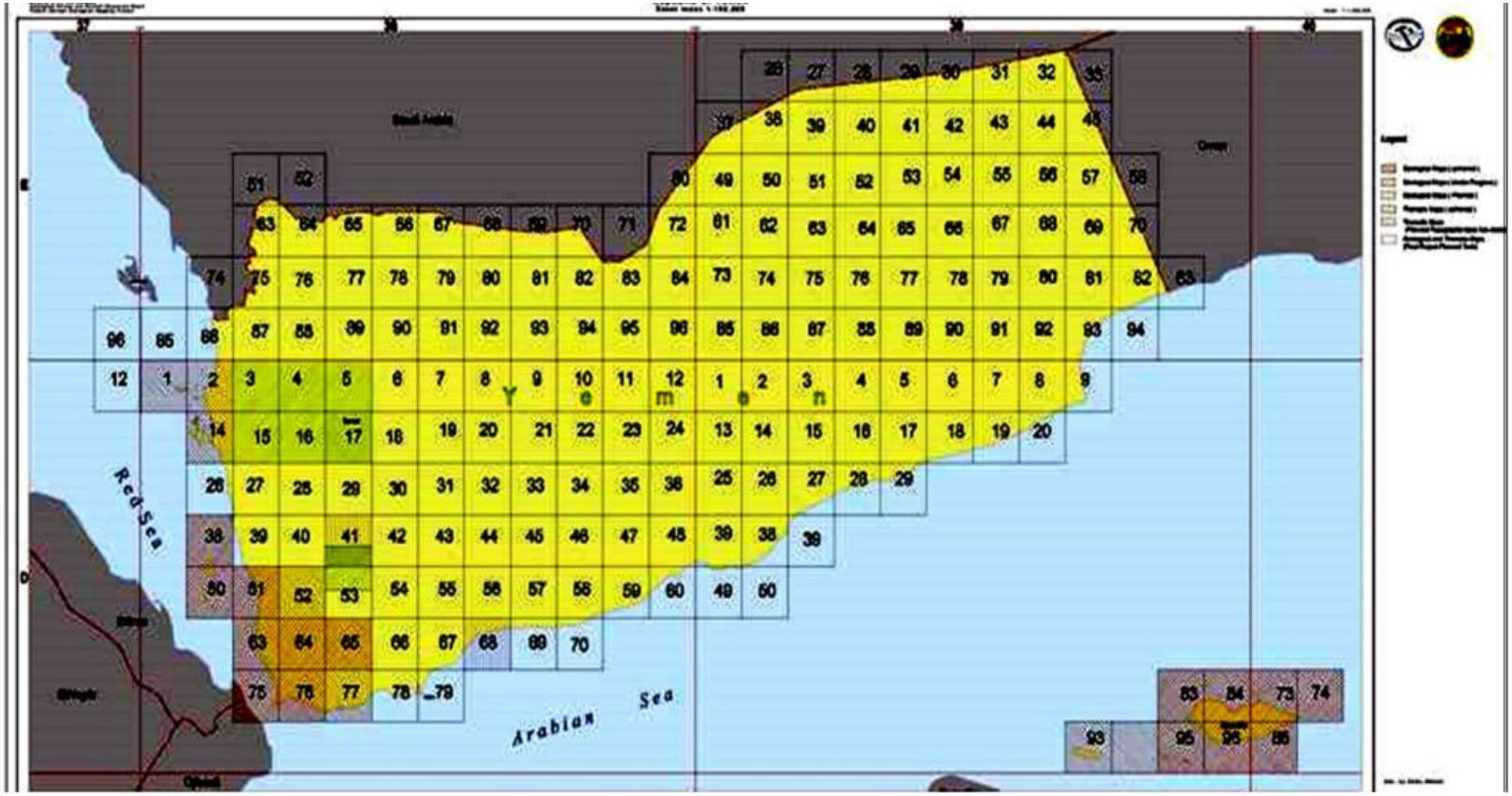
Minerals in Yemen

- **Geological map zonation**
  - **Sheets arrangement;**



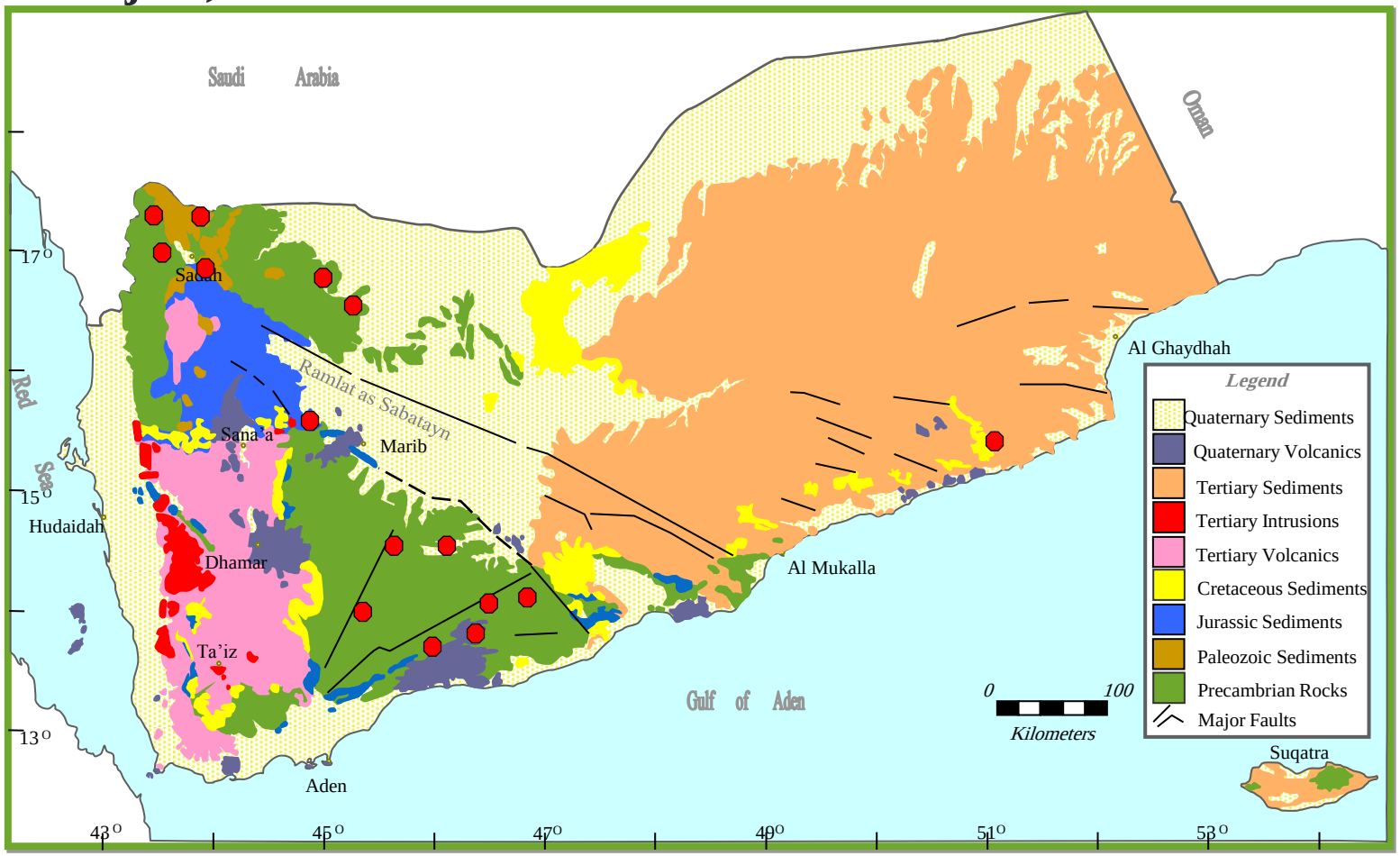
Minerals in Yemen

- **Topographic maps in Yemen**
  - **Sheets arrangement;**



Minerals in Yemen

- **Distribution of Old mining**
  - **Gold: at Aljaof, Sa'adah and Shabwah**



Minerals in Yemen

- **Yemen rich in minerals,** ؟
  - **The minerals which are recognized;**
  - **Metallic:**
    - **Gold, silver, mercury, platinum, nickel, cupper, zinc, cobalt, lead.**
  - **Non metallic:**
    - **Gypsum, carbonate, silica, halite, zeolite, feldspar.**
  - **Industrial stone:**
    - **Marble, limestone, sandstone, granite, travertine, tuff, piumus, scoria, ignimbrite.**

Minerals in Yemen

- **Infra structure;** ؟
  - **Roads; 14,600 kilometer asphaltic, 62,000 paved (2009).**
  - **Telecommunication**
  - **Airports; 5 international, 8 national.**
  - **Harbors; 6 international, 8 national.**

  - **Mining and quarry low, No. 22, 2010**

Minerals in Yemen

- **Mining and quarry low, No. 22, 2010**
- أنواع التراخيص المعدنية وفقاً لقانون المناجم والمحاجر رقم (٢٢) لسنة ٢٠١٠م

؟

نوع الترخيص	المساحة (كم٢)	المدة (سنة)	التمديد (سنة)	نوع الموارد المعدنية
استطلاع	10,000	1	سنة واحدة	المعادن الفلزية
	١ - ١٠		سنة واحدة	خامات المحاجر
استكشاف	2,000	٤	فترتين مدة كل فترة ٤ سنوات	المعادن الفلزية
	٥	٢	فترة واحدة مدتها سنتان	خامات المحاجر
تعدين	٠,٢٥ - ٢٠	25	فترات غير محدودة مدة كل فترة 10 سنوات	المعادن الفلزية
محجر	٠,٠٥ - ٢	10	فترات غير محدودة مدة كل فترة 10 سنة	خامات المحاجر
تعدين حرفي	لا تزيد عن 1	2	فترات غير محدودة مدة كل فترة سنة واحدة	المعادن الفلزية

Minerals in Yemen

- **Metallic minerals:** ؟
- **Occurrence and its geology;** ؟
  - **Gold;**
  - Wadi Madan Hadramawt: 678000 ton, 15ppm.
  - Alharegah; (Hajjah); 16000000 ton, 1.65ppm
  - Thamar, Sa’adah, Aljof, Albaida’a and Shabwah.
  - **Within basement**; hidan, Sagain, Wadi Riag, Wadi Alnmasah, Almatammah, Wadi Flhan, Bahrah, Kholan, Soog Aahim, Kasher, Washhah, Haradh, Ya’alan, Amsorrah.
  - **Within volcanic rocks**; Shaharah, Aatheen, Waragah, Othmah, Alwazeaiah.

Minerals in Yemen

- **Metallic minerals:**
- **Occurrence and its geology;**
  - **Gold;**

HAJJAH

HADRAMOUT

Minerals in Yemen

- **Metallic minerals:** ؟
- **Occurrence and its geology;**
  - **Zinc and lead;**
  - Related to Ramlat Alssabatain basin as a veins within limestone and dolomitic rocks.
  - Locations;
    - Aljabali area- Sana’a, 12.6 million ton , 9% zinc, and 1.2% lead, 68ppm Silver.
    - Tabag area- Shabwah, 12% zinc, and 3.8% lead.
    - Theebain area – Amran, 16% Zinc.
    - Rmah – Shabwah 7% zinc.
    - Wadi Ageeg, 1.5% zinc.

Minerals in Yemen

- **Metallic minerals:**
- **Occurrence and its geology;**
  - **Cupper, Nickel and cobalt;**
  - Within basement rocks;
  - Locations;
    - Soar area - Hajjah
    - Almasna’ah area- Sa’adah
    - Wadi Baihan – Shabwah
    - Alhamorah – Ta’aiz
    - Alma’aden Albaidah
    - Wadi Sarian Albaidah
    - Wadi Salabah Sa’adah

Minerals in Yemen

- **Metallic minerals:**
- **Occurrence and its geology;**
  - **Ferric, Titanium, and Vanadium;**
  - Within intermediate volcanic rocks interbedded within marble, and associated with basic and ultrabasic intrusive rocks.;
  - Locations;
    - Althaniaa – Ma’areb
    - Mokiras – Abian
    - Sa’adah
    - Sebian- Albaidah
    - Mawarem – Abian

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - Gypsum
  - Rock salt
  - Pure limestone
  - Magnesite and dolomite
  - Pure sandstone
  - Quartz
  - Feldspar
  - Natural zeolite
  - Perlite and pumice
  - Scoria
  - Heavy minerals
  - Celestine
  - Clays
  - Talc
  - Barite
  - Fluorite
  - Wollastonite
  - Pyrophyllite
  - Aggregate
  - Building and decorative stone

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Gypsum</u>**
  - In upper Jurassic at Shabwah
  - In upper Jurassic and lower Cretaceous at Sana'a and Marib.
  - In lower Eocene and Oligocene ages as a huge deposits at Almaharah, Abyan, Shabwah, and Hadramawt.
  - In middle-upper Miocene deposits gypsum alternated with salt and clay beds at Al-Hodaidah.

  - **Estimated Reserve**; 4.6 billion cubic meters.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Rock salt</u>**
  - Open pit mining operation is back to date at least to 1538.
  - As a salt domes within a Sab'atayn formation (Amran group) upper Jurassic, at Marib (Safir area), Shabwah (Ayad and Kharwa areas), and Hadramawt (Al-Mintaq area 100km north west of Al-Mukalla city)
  - Within AsSalif formation (Tihamah group) late tertiary age, at Al-Hodaidah (AsSalif area).
  - **Estimated Reserve;** 390 million cubic meters.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Pure limestone</u>**
  - There are 44 reported pure limestone occurrence in Yemen.
  - Their chemical properties and whitenence make them suitable for production of pure calcium carbonate, white cement, iron and steel, glass, paper and agriculture uses.
  - In Amran, Hadramawt and Shihr groups,
  - At Al-Maharah, Hadramawt Abyan, Amran, Sana'a, Lahj, and Shabwah Zones
  - **Estimated Reserve;** 3.6 billion cubic meters.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Magnesite and dolomite</u>**
  - They are one of the most important industrial minerals in Yemen.
  - Have high quality. In Precambrian, Jurassic and Tertiary age.
  - **Magnesite**
  - At Marib zone (Al-Thonyyah Area)
  - **Dolomite**
  - At Shabwah (near paved highways), Hadramawt (near paved highways within fuwwah formation), and
  - Al-Maharah Zone, (within Umm er Radhuma and Habshiyah formations (Hadramawt group))
  - **Estimated Reserve**; 58 million tons for magnesite, and 3.4 billion cubic meters for dolomite.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Pure sandstone</u>**
  - Suitable as raw material for glass manufacture.
  - There are 25 reported occurrence in Yemen.
  - In Wajid, Kuhlan formations and a Taweilah group.
  - Sana'a zone, at Thaqban area, Jabal Al-Nuqs area and Jabal Arishah area.
  - Shabwah Zone, at Habban area.
  - Sadah zone, at Al-habat area, Bani Uwair, Jabal Barash, Jabal Delan, Jabal Tolumos and Jabal Ablah areas.
  - Hajjah zone, at wadi Harad area.
  - Ta'iz zone, at Jabal Almansurah area.
  - Lahj zone, Saylat Bilah area.
  - **Estimated Reserve**; 2 billion cubic meters.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Quartz</u>**
  - Characterized by a high silica content.
  - Within Precambrian pegmatite, gneiss, and schistose rocks.

  - Hajjah zone, at Harad area
  - Sa’dah zone, at Al-minzlah and Jabal Uwaidah areas.

  - **Estimated Reserve**; 10 million cubic meters.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Feldspar**
  - There are 25 reported occurrence in Yemen.
  - As pegmatite lenses, within Precambrian basement rocks.

  - Hajjah zone, at Bawhal, Hyran, Bani Nashr and Sharis Wadis.
  - Abyan zone, at Sabib and Barhah wadis.
  - Shabwah zone, at Jibal Irqah, wadis of Al-Hanak and Abdan.
  - Remah zone, at Al-Salfyah area.

  - **Estimated Reserve**; 44 million cubic meters.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Natural zeolite</u>**
  - Occurred in a good quality in altered volcanic tuff.
  - In Ta’iz (Al-Adnah and Al-Ahyuq areas), Ibb (AL-Qa’idah area) and Thamar (Marayah area).
  - **Estimated Reserve**; 200 million cubic meters.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Perlite and pumice**
  - Are common in Cenozoic volcanic rocks of Yemen.
  - Perlitic rocks are found as acidic flows from a few to tens meter thickness.

  - **Perlite**
  - Thamar (Aanis) , Ibb, Lahj (Am-Shat), and Ta'iz (Al-Waze'yyah) areas.
  - **Pumice**
  - Thamar (Al-lisi area).
  -
  - **Estimated Reserve**; 1billion cubic meters for pumice and 353 million cubic meters for perlite.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Scoria</u>**
  - Their physical and chemical properties make them suitable as active mineral additive in Portland pozzolan cement production, and aggregates for light weight concrete.
  - Pyroclastic cones within the basaltic lava in the quaternary volcanic, are found at the five volcanic fields,
  - Sana'a- Amran, Thamar – Rada', Marib- Sirwah, Shuqrah and Balhaf-Bir Ali.
  - **Estimated Reserve**; 1.8 billion cubic meters .

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Heavy minerals</u>**
  - There are 15 heavy minerals-bearing beach sands deposits and occurrence in Yemen, containing ilmenite, rutile, zircon, magnetite, monazite, leucoxene, along the coastal plains.
  - Placer deposits formed in beach environments by concentrating due to the specific gravity of the mineral grains.
  - Arabian sea coast; (west of Al-Mukalla, and east Sifal beach, Raydat area, Sayhut area (), Al-Ghaydah area, Qishn area.
  - Red sea, AsSalif, AtTifaf, Al-Urj and Wadi Rasyan

  - **Estimated Reserve;** 500 million tons.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Celestine**
  - Mainly found in shales within Jize formation tertiary age. Its concentration in eastern part of Yemen.
  - Located on Hadramawt plateau, 70km north of Mukalla city. Occupy a surface of about 200km^2 , fossils are preserved by Celestine in nodular limestone (Umm er Radhoma formation).
  - Celestine-bearing beds vary in thickness from 1 to 8 meters, contain less than 1% Celestine, locally grade 5% Celestine.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Clays**
  - Widespread, occur in most area of Yemen.
  - Sana’a (Semictite at Khawlan and Sana’a, Belad Al-Rus),
  - Sa’dah (Kaolinite and Halloysite at Madak and Al Ain, Semictite at Sa’dah basin),
  - Al-Jawf (Barat),
  - Hadramawt within Jiza formation (montmorillonite at Ad Dhali’ah, Thilah AsSufla and Al-Shu’bah),
  - Thamar (Halloysite at Bakhran, Semictite at Bab AL-Filak) ,
  - Ibb (Semictite near the city and Wadi Rafud), Al-Hudaidah (Hays),
  - Abyan, and
  - Lahj (Semictite as loam near Al-Hawtah town and Lakidah) areas.

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Clays**
  - **Shale and Marl**; at
  - Sa’dah (Wadi Akbarah), and
  - Sana’a (Khulqah and Al-Mahajir).

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Talc**
  - Is derived from alteration of ophiolitic ultramafic rocks contained within metamorphic rocks.
  - Sa’dah (Jabal Al-Tohle)
  - Marib (Harib)
  - Al Baydah (Sabah)
  - Abyan (Lawder)

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Barite**
  - There are 13 reported occurrence in Yemen.
  - Al-Masilah, Al-Jubah, Lawder, Al-Jabali

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Fluorite**
  - Associated with Precambrian basement rocks and tertiary trap series.
  - Sabwah (Irqah)
  - Abyan (Nigd Emsani)

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Wollastonite</u>**

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **Pyrophyllite**

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Aggregate</u>**
  - They are wide distribution in Yemen with low cost production.
  - There are 280 aggregate quarries with a total production of more than 14 million tons.
  - Most of it are produced from hard rock formations or from naturally occurring deposits such as sand and gravel .
  - The most important sources of aggregates in Yemen are sandstone, basalt, limestone, pumice scoria, igneous and metamorphic rock, sand and gravel.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Aggregate</u>**
  - **Sandstone**; (provide about 36% of the aggregate product) .
  - Small quarries and pits in Sana'a and Amran, from cretaceous sandstone of Tawillah grope.

  - **Basalt**; (provide about 34% of the aggregate product).
  - Predominantly concentrated in Sana'a, Thamar, Aden , Ibb, Ta'iz, Lahg, Ad Dalli' and Al-Hudidah .
  - The major source are the basalt of tertiary age.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Aggregate</u>**
  - **Pumice**; (provide about 16% of the aggregate product) .
  - Occur east of Thamar. 1billion m^3.
  - Are mined from pits using end loaders.

  - **Limestone**; (7% of the aggregate product) exploited to obtain aggregate (broken stone), crushed rocks are also used as aggregate, and to a lesser extend in the manufactures of terrazzo tiles.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Aggregate</u>**
  - **Scoria**; (provide about 3% of the aggregate product) .
  - Lightweight aggregate to produce lightweight concrete and lightweight blocks. Recently used in concrete tilt panels.
  - Pyroclastic cones within the basaltic lava in the quaternary volcanic.

  - **Igneous and metamorphic**; (provide small% of the aggregate product) .
  - Predominantly concentrated in Sa’dah, Hajjah, Aljawf, Marib, AlBayda, Hadramawt, Ta’iz, and Shabwah.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Aggregate</u>**

  - **Sand and gravel**; (provide about 4% of the aggregate product) .
  - Is composed of durable particles and rich in silica (quartz, quartzite and flint).
  - Produced from small quarries and pits in Hadramawt, Ad Dalli', Lahj, Marib, AlMahrah and Sa'dah.

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Building and decorative (ornamental) stone</u>**
  - **Tuff and ignimbrite**;
    - Occurrence concentrated in the western highlands of Yemen , in Sana'a, Thamar, Ta'iz and Ibb. Used as decorative stone.
    - Estimated Reserve; 343million cubic meter.
  - **Basalt**;
    - Compact and vesicular basalts used as building and decorative stone.
    - Occurrence concentrated in the western highlands of Yemen , in Sana'a, Thamar, Ta'iz and Ibb.
    - Estimated Reserve; 142 million cubic meter

Minerals in Yemen

- **Industrial minerals;** ؟
- **Occurrence and its geology;**
  - **<u>Building and decorative (ornamental) stone</u>**
  - **Limestone**
    - Used as building and decorative stone.
    - Occurrence concentrated in Sana'a, Amran,Sa'dah, Marib, Hadramawt, Abyan, Lahj, Shabwah , AlMahrah, and Ta'iz.
    - Estimated Reserve; 5.3 billion cubic meter
  - **Travertine**
    - Occurrence concentrated in Sana'a, Ibb and Al-Dali'.
    - Estimated Reserve; 2.5 million cubic meter

Minerals in Yemen

- **Industrial minerals;**
- **Occurrence and its geology;**
  - **<u>Building and decorative (ornamental) stone</u>**
  - **Granite and gabbro**
    - Precambrian granite and Tertiary granite.
    - Occurrence concentrated in Albaidah, Al-jawf, Marib, Shabwah, Sa'adah, Abyan, Hajjah, Al-hudaidah, and Lahj
    - Estimated Reserve; 2.7 billion cubic meter
  - **Marble**
    - The first type result from regional metamorphism of Jurassic limestone, in contact with volcanic rocks, the second type outcrops as large bodies in Precambrian rocks.
    - Occurrence concentrated in Sana'a, Marib, Ta'iz, Hajjah, Shabwah, and Abyan. Estimated Reserve; 2.7 billion cubic meter

Minerals in Yemen

- **Investments (exploitations) Opportunities in Yemen;** ؟
  - The extent and variety of the mineral recorded; Metallic, Non metallic, Industrial stone.
  - Infra structure; Roads, Telecommunication, Airports, Harbors. ؟
  - Mining and quarrying low, No. 22, 2010
  - The exceptional geological environment of Yemen in the region.
  - The geographical situation of Yemen in the world.

Minerals in Yemen

Thank you for your attention